

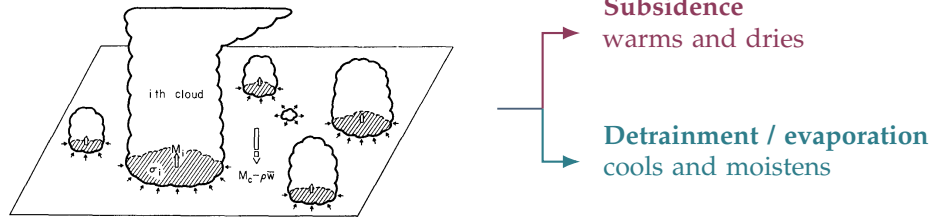
Arakawa–Schubert (1974): cloud feedbacks

Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I, Arakawa and Schubert (1974), J. Atmos. Sci. 31, 674–701. The theory predicts the collective effect of cumulus clouds and their response to a slowly evolving environment, including the subcloud mixed layer.

Cloud i covers the area fraction σ_i and carries the mass flux $\rho\sigma_i w_i$. The sum M_c drives compensating environmental subsidence. In a stably stratified atmosphere this warms and dries the environment. Detrainment supplies cloud air and liquid water, whose evaporation cools and moistens. The diagram labels the two corresponding terms in the original environmental budgets.

The variables are $s = c_p T + gz$ and vapour mixing ratio q . Hats denote detraining-air values, the star denotes saturation, $\hat{l}$ is detrained liquid water, and L is latent heat. For legibility the slide groups the original large-scale terms as $LS_s = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{s} - \rho \bar{w} \partial_z \bar{s} + \bar{Q}_R$ and $LS_q = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{q} - \rho \bar{w} \partial_z \bar{q}$.

The equations assume a small active cloud fraction and a horizontally uniform environment. Figure 1 is the paper's conceptual mass-exchange sketch, not a measured horizontal cloud map. Source: Sections 2–3, Figure 1, Equations (7), (74)–(75).



$$M_c = \sum_i \rho \sigma_i w_i$$

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + LS_s$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + LS_q$$

Mass flux and detrainment set the environmental heating and moistening

Cloud type λ : fractional entrainment.

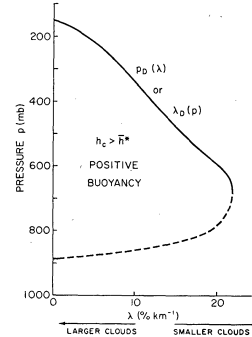
$$\partial_z \eta = \lambda \eta \quad \Rightarrow \quad \eta = e^{\lambda(z-z_B)}$$

$$\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$$

h : moist static energy. h_M : mixed-layer value.

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

$$h_c(z_D) = \hat{h}^* \approx \bar{h}^*$$



More entrainment gives a lower cloud top in the reference sounding

Entrainment determines the plume structure

The ensemble is divided into entraining plumes with fractional entrainment rate λ , assumed constant with height. The normalized mass flux satisfies $\partial_z \eta = \lambda \eta$ and $\eta(z_B) = 1$, so it grows exponentially up to the plume top. The mass-flux reference level z_B is the mixed-layer top, which can lie below cloud base.

The moist static energy $h = s + Lq$ follows the entraining-plume budget. On the slide, the arguments are suppressed: $\eta(z, \lambda) h_c(z, \lambda) = h_M + \lambda \int_{z_B}^z \eta(z', \lambda) \bar{h}(z') dz'$. Entrainment dilutes mixed-layer air with environmental air. The cloud top z_D is the level of vanishing buoyancy, $h_c(z_D) = \hat{h}^*(z_D)$, approximately $\bar{h}^*(z_D)$. Figure 8 shows the resulting cloud-top pressure for the reference tropical sounding. Follow the solid branch: increasing entrainment gives a lower top, hence higher pressure.

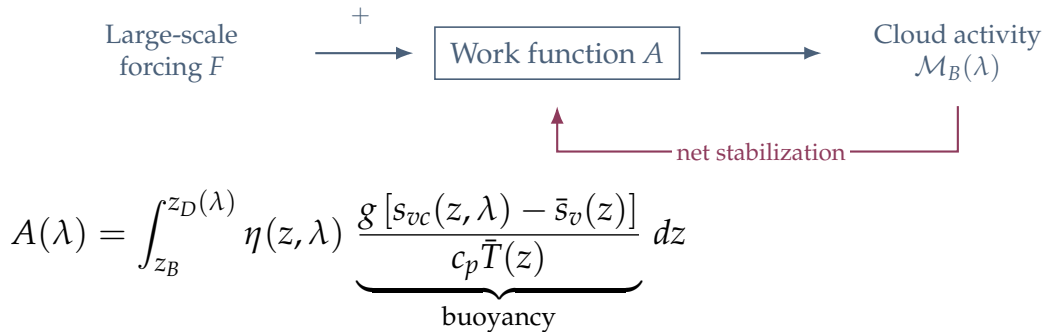
The integral for $M_c(z)$ sums the plume types still present at height z . Its upper limit $\lambda_D(z)$ selects types whose top reaches that level. The remaining unknown is the mass-flux density $\mathcal{M}_B(\lambda)$, not a number or radius density. Source: Section 4, Equations (78)–(88), (92), (95)–(97), Figure 8, and Section 5 for the mixed-layer boundary values.

Cloud work function and quasi-equilibrium

The cloud work function is the buoyancy integral weighted by normalized mass flux. It measures kinetic-energy generation per unit mass flux at the reference level: the kinetic-energy budget is $dK/dt = A\mathcal{M}_B - \mathcal{D}$. The environment determines A , including the mixed-layer contribution. Positive A permits buoyant energy production but does not by itself prescribe cloud counts or triggering.

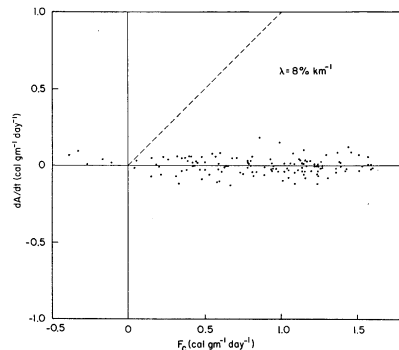
The map shows the feedback: large-scale forcing changes A , buoyancy supports cloud activity, and the ensemble modifies the environment and thus its own work function. Net stabilization dominates the adjustment. Individual entries of K need not all be negative, since detrainment by shallow clouds can help deeper clouds.

The tendency equation separates the cloud contribution, $\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'$, from the large-scale forcing F . Under quasi-equilibrium the tendency is small compared with these two contributions, giving the integral equation that closes the mass-flux spectrum. The equality holds for active cloud types, with a nonnegative spectrum and an inequality for absent types. This is the principal theoretical result of Part I. It does not impose a steady temperature or moisture profile. Source: Sections 6–7, Equations (132)–(133), (142), (150) and (158). The map summarizes these equations.



$$\frac{dA}{dt} = \int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0 \quad \mathcal{M}_B(\lambda) \geq 0$$

Fast cloud adjustment balances the large-scale forcing



Marshall Islands. Dashed line: $dA/dt = F_C$.

Timescale separation

$$\tau_{\text{adj}} \sim 10^3\text{--}10^4 \text{ s}$$

$$\tau_{\text{LS}} \gtrsim 10^5 \text{ s}$$

$$\text{Observed: } |dA/dt| \ll |F_C|$$

Clouds nearly balance the forcing.

The observed work function changes much less than the forcing alone predicts

Observations support the adjustment balance

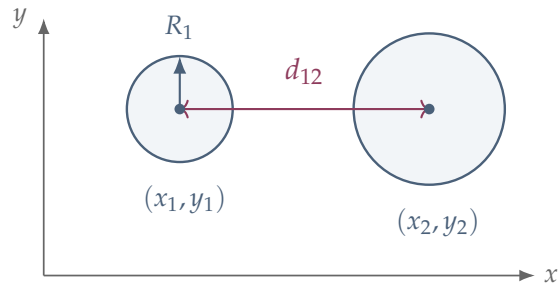
The paper estimates a cloud adjustment time of $10^3\text{--}10^4 \text{ s}$ and a large-scale forcing time of at least 10^5 s . This separation motivates dropping the relatively small work-function tendency in the closure equation. The estimates are orders of magnitude, not measured lifetimes of individual thermals.

Figure 13 compares the observed tendency of the cloud work function with cloud-layer forcing F_C from the Marshall Islands data of Yanai et al. (1973). The upper panel, reproduced here, uses $\lambda = 8\% \text{ km}^{-1}$. Without cloud adjustment, the observations would lie along the dashed line $dA/dt = F_C$. Instead, the tendency is generally much smaller than the forcing, which reaches about $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$. The lower panel for $\lambda = 16\% \text{ km}^{-1}$ gives the same qualitative result.

The test supports quasi-equilibrium in this setting. It omits the mixed-layer forcing F_M , which could not be evaluated from the conventional observations. Figure 14 separately shows that temperature and moisture profiles evolve while their combined work function remains small. Numerical solution methods and applications of the closure are deferred to Part II. Source: Section 7, Equations (146)–(154), Figures 13–14, and Section 8.

Formulated: mass flux by cloud type, $\mathcal{M}_B(\lambda)$.

Spatial quantities (schematic)



Not reported for thermals

Size distribution R_i

Positions (x_i, y_i)

Distances d_{ij}

A mass-flux spectrum leaves the horizontal thermal field unspecified

Scope: §4 and §7. The sketch defines the quantities sought. It represents no observed or simulated thermal field.

No 2D distribution of thermals

The target of our reading was the distribution of thermals in a horizontal plane. The sketch defines sizes, positions and centre-to-centre separations. It is illustrative only. Arakawa and Schubert do not report distributions of these thermal quantities.

The model resolves cloud types through their fractional entrainment and vertical structure, while closing their collective mass-flux contribution. A spectrum over λ is not a two-dimensional spatial distribution. The paper explicitly separates mass flux from cloud number, lifetime and single-cloud transport, which are not determined independently. Its use of a relation between entrainment and cloud radius does not yield a measured thermal-radius distribution.

Horizontal uniformity of the environment is a modelling assumption, not evidence for randomly or uniformly positioned thermal centres. The cloud sketch in Figure 1 likewise supplies no spatial statistics. Source: Sections 4 and 7, particularly the discussion of Equations (94) and (145).